

NOTES ON RATIONAL CHOICE UNDER UNCERTAINTY

John Duggan
May 20, 2010

Contents

1	Expected Utility	2
2	Uniqueness of the vN-M Representation	8
3	States of the World	11
4	The One-dimensional Choice Model	15
5	Stochastic Dominance	19
6	Measures of Risk Aversion	22
7	The Multidimensional Choice Model	26
8	Mean-Variance Analysis	26
9	Experimental Evidence	28
10	The Paradox of Voting	29

We consider a decision maker whose welfare is determined by the outcome realized from a set A of alternatives, and we allow for the possibility that there is uncertainty about what the outcome in A will be. For example,

- A may consist of the possible winners of an election, candidate A or candidate B , and a candidate who adopts a particular campaign platform may not know who the winner will be;
- A may consist of various possible economic outcomes, and a politician may not know the economic effects of different public policies;

- A may consist of two types of politician, conservative and liberal, and voters may not know for sure what type a candidate is;
- A may consist of the possible winners of an election, candidates A or B , and a voter who votes for A (or B) may not know who the winner will be.

These are all examples of decision problems under uncertainty: What campaign platform should a candidate adopt? Which policy should a politician choose? Which candidate should a voter vote for? Should a voter bother to vote at all? To answer these questions, we first need a way of modelling uncertainty.

1 Expected Utility

Number the alternatives in A as $\{a_1, \dots, a_m\}$. We represent uncertainty about the outcome from A by a probability distribution, a vector $\lambda = (\lambda_1, \dots, \lambda_m)$, which tells us the probability of alternative a_j is λ_j . Of course, $\lambda_j \geq 0$ for all j , and $\sum_{j=1}^m \lambda_j = 1$. In choice theory, we usually call λ a *lottery*. Let Λ denote the set of all lotteries, i.e.,

$$\Lambda = \left\{ \lambda \in \mathbb{R}_+^m \mid \sum_{j=1}^m \lambda_j = 1 \right\}.$$

In other words, Λ is the “unit simplex” in $\mathbb{R}^m$. The simplex has m vertices, $\delta_1, \dots, \delta_m$, where $\delta_j = (0, \dots, 0, 1, 0, \dots, 0)$ is a vector with a one in the j th coordinate, zeros elsewhere. This is the lottery that puts probability one on alternative a_j , and for this reason we refer to these lotteries as *degenerate*.

In the above examples, a decision maker needs to make a choice that will determine some lottery. To understand how that choice will be or should be made, we need to consider preferences over lotteries. We assume that the decision maker is an individual and has a strict preference relation P on Λ , with weak preference R and indifference I derived as in the standard rational choice model: given $\lambda, \mu \in \Lambda$, $\lambda R \mu$ if and only if not $\mu P \lambda$, and $\lambda I \mu$ if and only if $\lambda R \mu$ and $\mu R \lambda$. We will investigate the most prominent model of choice under uncertainty, which is defined by the following primitives.

Expected Utility Model:

A	alternatives, $a_1, \dots, a_m$
Λ	lotteries on A
P	ordering on Λ

We imagine that the decision maker is given a choice among a set of $\Phi \subseteq \Lambda$ of feasible lotteries, and that a lottery λ is a viable choice if it is *maximal*, i.e., for all $\mu \in \Phi$, $\lambda R \mu$.

Do all orderings of lotteries make sense? Or can we rule some out on first principles? One sensible restriction is that if there is one lottery λ that is strictly preferred to all others, then it should be degenerate. Otherwise, λ puts positive probability on at least two alternatives, and one of those, say a_j , will be at least as good as the other, say a_k . But then if we move probability from a_k to a_j , we should have a lottery that is at least as good as λ , contradicting the supposition that λ is uniquely best.

To discuss further restrictions, it is useful, given two lotteries $\lambda, \mu \in \Lambda$ and $\alpha \in [0, 1]$, to define the new lottery $\alpha\lambda + (1 - \alpha)\mu$. Writing this out in vector form,

$$(\alpha\lambda_1 + (1 - \alpha)\mu_1, \dots, \alpha\lambda_m + (1 - \alpha)\mu_m),$$

this is just the lottery that puts probability $\alpha\lambda_j + (1 - \alpha)\mu_j$ on each alternative a_j . We may refer to this as a “mixture” of λ and μ . (You can check that these probabilities are non-negative and sum to one.) What is the intuitive meaning of the new lottery? Imagine a two-stage lottery, where we randomly decide whether to use lottery λ or lottery μ (with probability α on λ and $1 - \alpha$ on μ), and we then draw the final outcome in A according to whichever lottery was chosen in the first stage of randomization. The probability distribution on A determined by this two-stage lottery is just $\alpha\lambda + (1 - \alpha)\mu$. Geometrically, of course, the mixed lottery is just a convex combination of the original ones.

Returning to the question of preference restrictions, the following logic is appealing. Let λ and μ be two lotteries such that $\lambda P \mu$, and consider the mixture $\nu = \alpha\lambda + (1 - \alpha)\mu$, where $\alpha \in (0, 1)$. Then it stands to reason that $\nu P \mu$. Why? Viewed as a compound lottery, ν is equivalent to choosing μ with probability $1 - \alpha$ and a better lottery λ with positive probability $\alpha > 0$; the worst possibility is that we randomize over alternatives according to μ , but at least there is a positive probability that we randomize using the superior lottery. Thus, it seems reasonable to assume that preferences over lotteries possess some natural convexity properties. In fact, this extends the intuition from the above example in which one lottery was uniquely best.

Are there other restrictions we can impose? The literature on decision theory has considered many restrictions, formulated as axioms on the ordering P . A very weak axiom is that the ordering P has no “gaps.”

Intermediate values For all $\lambda, \lambda', \lambda''$ with $\lambda P \lambda' P \lambda''$, there exists $\alpha \in (0, 1)$ such that $\lambda' I (\alpha\lambda + (1 - \alpha)\lambda'')$.

The latter axiom is implied if, for example, the decision maker’s ordering has a continuous utility representation. Another common axiom used is next.

Independence For all $\lambda, \lambda', \lambda''$ such that $\lambda P \lambda'$, and for all $\alpha \in (0, 1)$, we have

$$(\alpha\lambda + (1 - \alpha)\lambda'') P (\alpha\lambda' + (1 - \alpha)\lambda'').$$

Independence is also referred to as the “Substitution” axiom. The intuitive justification for this axiom is to consider choosing between $\mu = \alpha\lambda + (1 - \alpha)\lambda''$ and $\mu' = \alpha\lambda' + (1 - \alpha)\lambda''$; and imagine an initial randomization over λ (with probability α) and, depending on whether μ or μ' is chosen, λ or λ' (with probability $1 - \alpha$). In case λ is realized in the initial randomization, the choice between μ and μ' is moot; the choices only differ when λ or λ' is realized, in which case, because $\lambda P \lambda'$, μ should be superior.

While the axioms of Independence and Intermediate values are simple and seem quite reasonable, it turns out that in conjunction, they are very restrictive. In fact, they imply that preferences over lotteries have a specific—and very convenient—functional form, known as “expected utility.” The following characterization is due to von Neumann and Morgenstern (1944). The theorem provides the theoretical foundation for most formal analyses of choice under uncertainty.

Theorem 1.1 (von Neumann and Morgenstern). *The strict preference relation P satisfies Independence and Intermediate values if and only if there is a function $u: A \rightarrow \mathbb{R}$ such that, for all $\lambda, \mu \in \Lambda$,*

$$\lambda P \mu \Leftrightarrow \sum_{j=1}^m \lambda_j u(a_j) > \sum_{j=1}^m \mu_j u(a_j). \quad (1)$$

We call any function satisfying (1) for all λ and μ a *von Neumann-Morgenstern representation*. We specifically avoid the term “utility,” as we elaborate at a later point. The quantity

$$U(\lambda) = \sum_{j=1}^m \lambda_j u(a_j)$$

is called the *expected utility* of lottery λ , and you can see that the function U is really an ordinal utility representation of the decision maker’s preferences over lotteries: condition (1) is equivalent to

$$U(\lambda) > U(\mu) \Leftrightarrow \lambda P \mu,$$

as required for an ordinal utility representation. Of course, the condition in (1) is equivalent to the following: for all $\lambda, \mu \in \Lambda$,

$$\lambda R \mu \Leftrightarrow U(\lambda) \geq U(\mu).$$

Both conditions imply that for all $\lambda, \mu \in \Lambda$, the decision maker is indifferent between λ and μ if and only if $U(\lambda) = U(\mu)$.

What do the indifference curves of P look like in Λ ? Of course, the indifference curves are the level sets of U . Rewriting the expression for expected utility as

$$U(\lambda) = \lambda \cdot (u(a_1), \dots, u(a_m)),$$

note that U is a linear function of λ with gradient $(u(a_1), \dots, u(a_m))$, so its level sets are parallel lines (or planes or hyperplanes in higher dimensions). A monotonic transformation of U will also represent the decision maker's preferences over lotteries, but it may not have the nice expected utility form.

It is straightforward to verify that expected utility preferences satisfy the two axioms. We now assume P satisfies the axioms and deduce the expected utility form. We provide a simple proof of Theorem 1.1 for the case of three alternatives in which the corresponding degenerate lotteries are strictly ranked. We begin with two general lemmas. The first establishes a convexity property of preferences over lotteries. Specifically, suppose one lottery λ is strictly preferred to another λ' ; then every lottery on the line between the two that is closer to λ is strictly preferred to every lottery on the line closer to λ'' .

Lemma 1.2. *If $\lambda P \lambda'$, then for all $\alpha, \beta \in [0, 1]$ with $\alpha > \beta$, we have*

$$(\alpha\lambda + (1 - \alpha)\lambda')P(\beta\lambda + (1 - \beta)\lambda').$$

Proof. Denote $\tilde{\lambda} = \alpha\lambda + (1 - \alpha)\lambda'$, and note that Independence directly implies $\tilde{\lambda} P \lambda'$ (for this, we set $\lambda'' = \lambda'$ to apply the axiom). Denote $\hat{\lambda} = \beta\lambda + (1 - \beta)\lambda'$, and note that we may write

$$\hat{\lambda} = \gamma\tilde{\lambda} + (1 - \gamma)\lambda',$$

where $\gamma = \frac{\beta}{\alpha} \in (0, 1)$. Again, independence implies $\tilde{\lambda} P \hat{\lambda}$ (for this, we set $\lambda'' = \tilde{\lambda}$ to apply the axiom), as required. ■

The next lemma extends the Independence axiom to the case in which the decision maker is indifferent between two lotteries: indifference is maintained when we mix those lotteries with equal weight on any third lottery. In particular, if the decision maker is indifferent between two lotteries, then every lottery on the line between them is equally as good.

Lemma 1.3. *If $\lambda I \lambda'$, then for all $\alpha, \beta \in [0, 1]$, we have*

$$(\alpha\lambda + (1 - \alpha)\lambda')I(\beta\lambda + (1 - \beta)\lambda').$$

Proof. It is sufficient to show $(\beta\lambda + (1 - \beta)\lambda')I\lambda$, as an analogous argument shows $(\alpha\lambda + (1 - \alpha)\lambda')I\lambda$, and the lemma then follows from transitivity of indifference.

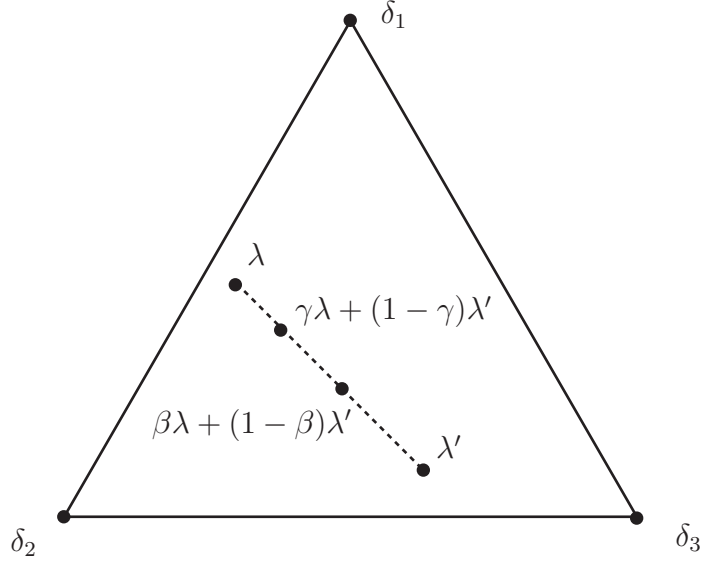


Figure 1: Proof of Lemma 1.3

Otherwise, there are two cases: $(\beta\lambda + (1 - \beta)\lambda')P\lambda$ or $\lambda P(\beta\lambda + (1 - \beta)\lambda')$. As the arguments are similar in the two cases, we consider only the first. Let $\gamma = \frac{1+\beta}{2} > \beta$, and note that Lemma 1.2 implies

$$(\beta\lambda + (1 - \beta)\lambda')P(\gamma\lambda + (1 - \gamma)\lambda')P\lambda. \quad (2)$$

See Figure 1. Furthermore, since $(\gamma\lambda + (1 - \gamma)\lambda')P\lambda I\lambda'$, we have

$$(\gamma\lambda + (1 - \gamma)\lambda')P\lambda'. \quad (3)$$

But with (3), Lemma 1.2 implies $(\gamma\lambda + (1 - \gamma)\lambda')P(\beta\lambda + (1 - \beta)\lambda')$, contradicting (2). $\blacksquare$

To complete the sketch of the proof, we focus on the simple case of three basic alternatives, and we assume that the degenerate lotteries corresponding to these alternatives are strictly ranked: $\delta_1 P \delta_2 P \delta_3$. By the Intermediate values axiom, we may choose an $\alpha \in (0, 1)$ and define $\tilde{\lambda} = \alpha\delta_1(1 - \alpha)\delta_3$ such that $\tilde{\lambda} I \delta_2$. By Lemma 1.3, the indifference “curve” between δ_2 and $\tilde{\lambda}$ is a straight line. See Figure 2. I claim Independence implies that all indifference “curves” are parallel to this line. Indeed, consider any $\beta \in (0, 1)$. Define

$$\begin{aligned} \lambda' &= \beta\tilde{\lambda} + (1 - \beta)\delta_1 \\ \lambda'' &= \beta\delta_2 + (1 - \beta)\delta_1. \end{aligned}$$

If, for example, $\lambda' P \lambda''$, then we can choose $\gamma \in (0, 1)$ such that $\gamma\lambda' + (1 - \gamma)\delta_3$ and $\gamma\lambda'' + (1 - \gamma)\delta_3$ belong to the line between δ_2 and $\tilde{\lambda}$, so these lotteries are both indifferent to δ_2 . See Figure 2. But Independence would imply

$$(\gamma\lambda' + (1 - \gamma)\delta_3)P(\gamma\lambda'' + (1 - \gamma)\delta_3),$$

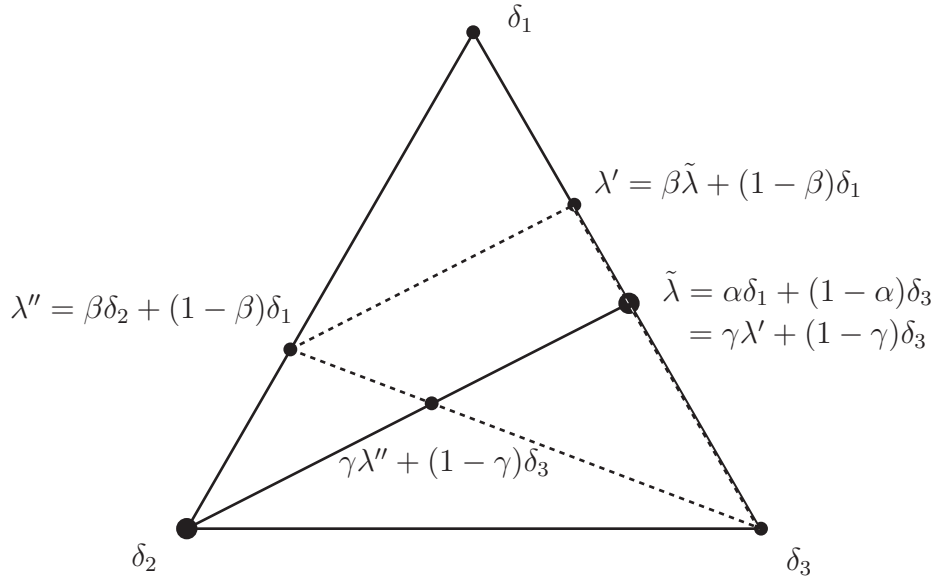


Figure 2: Parallel indifference curves, part 1

a contradiction. A similar contradiction rules out $\lambda''P\lambda'$, and therefore $\lambda'I\lambda''$. Then Lemma 1.3 implies that the indifference curve between λ' and λ'' is the straight line connecting the two lotteries. An analogous argument can be applied if we mix δ_2 and $\tilde{\lambda}$ with δ_3 , and we conclude that indifference curves are indeed parallel straight lines. See Figure 3.

Next, we exhibit a linear utility function over the set of lotteries that represents the preferences depicted in Figure 3. Define the linear function $U(\lambda) = \gamma \cdot \lambda$, where the fixed gradient is $\gamma = (1, \alpha, 0)$. (Technically speaking, the gradient depicted in the figure is the projection of γ onto the unit simplex.) In particular,

$$U(\delta_2) = U(0, 1, 0) = \alpha$$

and

$$U(\tilde{\lambda}) = U(\alpha\delta_1 + (1 - \alpha)\delta_3) = U(\alpha, 0, 1 - \alpha) = \alpha.$$

That is, U takes the same value at δ_2 and $\tilde{\lambda}$, so the line connecting these two lotteries lies on a level set of U . To verify that U represents the decision maker's preferences over lotteries, we need only check that it takes higher values on line segments closer to δ_1 , but this is immediate; it follows because $U(\delta_1) = 1 > 0 = U(\delta_3)$. We conclude that

$$u(a_1) = 1, u(a_2) = \alpha, u(a_3) = 0$$

is a von Neumann-Morgenstern representation for the decision maker, as required.

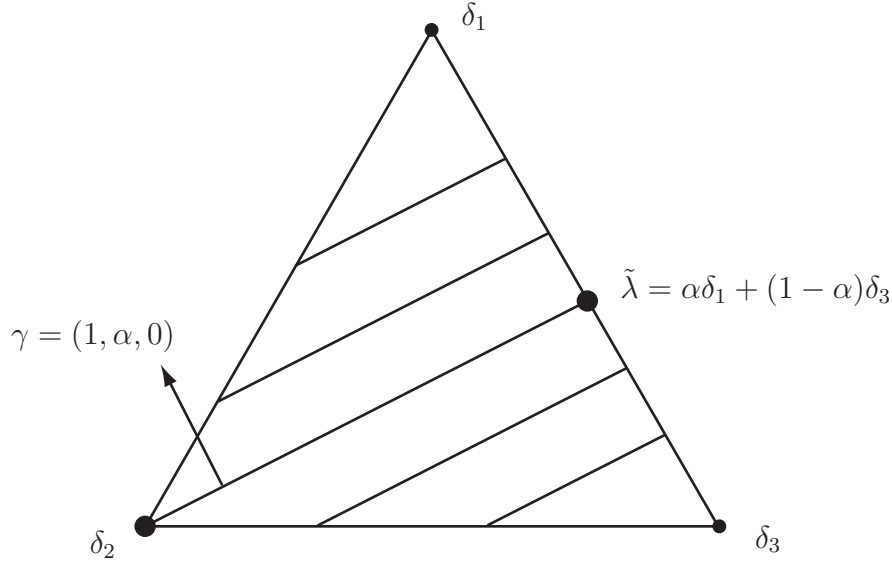


Figure 3: Parallel indifference curves, part 2

2 Uniqueness of the vN-M Representation

Since expected utility U , from Theorem 1.1, is an ordinal utility representation of the decision maker's preferences over lotteries, it follows that any strictly increasing transformation of U is an equally valid representation. In particular, a corollary of the theorem is that if preferences satisfy Independence and Intermediate values, then there is a mapping $v: A \rightarrow \mathbb{R}$ such that for all $\lambda, \mu \in \Lambda$,

$$\lambda P \mu \Leftrightarrow \prod_{j=1}^m v(a_j)^{\lambda_j} > \prod_{j=1}^m v(a_j)^{\mu_j},$$

giving us a multiplicative representation rather than an additive one. The proof is trivial: under the two axioms, Theorem 1.1 yields a von Neumann-Morgenstern utility u satisfying (1); since the exponential function is strictly increasing, it follows that the function $V: \Lambda \rightarrow \mathbb{R}$ defined by

$$V(\lambda) = e^{U(\lambda)} = \prod_j^m (e^{u(a_j)})^{\lambda_j}$$

is an ordinal utility representation; and then we can define v by $v(a_j) = e^{u(a_j)}$ to fulfill the claim.

We cannot, however, take arbitrary increasing transformations of the von Neumann-Morgenstern representation u without altering the preferences of the decision maker. While u is a function defined on elements of A , it is used to represent preferences

over *lotteries* on A . Of course, it determines preferences over A as well, because we can identify alternative a_j with the degenerate lottery δ_j . It then follows from (1) that a_j is weakly preferred to a_k if and only if $u(a_j) \geq u(a_k)$. If you take an arbitrary monotonic transformation of u , you will not change the implied ordering of A , but you *may* change the decision maker's preferences over lotteries.

To see this, suppose A contains three alternatives, and consider the von Neumann-Morgenstern utility function u and two monotonic transformations of it: $2u + 3$ and u^3 .

	u	$2u + 3$	u^3
x	1	5	1
y	.6	4.2	.2
z	0	3	0

Obviously, neither of these transformations of u has affected the implied ordering of A , but consider the decision maker's preference between y for sure, i.e., the lottery $(0, 1, 0)$, and the even chance lottery between x and z , i.e., $(.5, 0, .5)$. Under u and $2u + 3$, the former is preferred to the latter, but this preference is reversed for u^3 !

The lesson is that if we are using a von Neumann-Morgenstern representation function to represent preferences over lotteries, then we cannot subject it to arbitrary monotonic transformations. Transformations like $2u + 3$ (called "affine transformations") are okay, but not others. The next result formalizes these observations.

Theorem 2.1. *Let $u: A \rightarrow \mathbb{R}$ be a von Neumann-Morgenstern representation. Then $v: A \rightarrow \mathbb{R}$ is also a von Neumann-Morgenstern representation if and only if there exist constants $\alpha > 0$ and β such that $v = \alpha u + \beta$, i.e., for all $x \in A$, $v(x) = \alpha u(x) + \beta$.*

Proof. If u is a von Neumann-Morgenstern representation, if $\alpha > 0$, and if $v = \alpha u + \beta$, then it is easy to check that v also satisfies (1) for all λ and μ . For the other direction, suppose u and v satisfy (1) for all λ and μ but there do not exist $\alpha > 0$ and β such that $v = \alpha u + \beta$. In particular, note that u and v are not constant on A . Then there must exist $x, y, z, w \in A$ such that

$$\frac{u(x) - u(y)}{u(z) - u(w)} \neq \frac{v(x) - v(y)}{v(z) - v(w)}. \quad (4)$$

Otherwise, equality must hold instead, i.e.,

$$\frac{v(x) - v(y)}{u(x) - u(y)} = \frac{v(z) - v(w)}{u(z) - u(w)},$$

for all $x, y, z, w \in A$. Choose $c, d \in A$ arbitrarily such that $u(c) \neq u(d)$, and define $\alpha = (v(c) - v(d))/(u(c) - u(d))$. Note that $\alpha > 0$. (Why?) Then

$$v(x) - v(y) = \alpha(u(x) - u(y))$$

for all $x, y \in A$. Now choose $b \in A$ arbitrarily, define $\beta = v(b) - \alpha u(b)$, and note that

$$v(x) = \alpha u(x) + \beta$$

for all $x \in A$, a contradiction. Therefore, we may choose $a, b, c, d \in A$ such that (4) holds. I claim a consequence is that u and v do not both satisfy (1) for all lotteries λ and μ . To see this, let $\lambda = (1/m, \dots, 1/m)$ denote the even chance lottery over all alternatives, and define μ as follows. For $\epsilon > 0$ small, take probability ϵ from b and transfer it to a , and take probability

$$\epsilon \left(\frac{u(a) - u(b)}{u(c) - u(d)} \right)$$

from c and transfer it to d . Leave the probabilities of all other alternatives the same. Then

$$\begin{aligned} & \sum_{j=1}^m \lambda_j u(a_j) - \sum_{j=1}^m \mu_j u(a_j) \\ &= -\epsilon u(a) + \epsilon u(b) + \epsilon \left(\frac{u(a) - u(b)}{u(c) - u(d)} \right) u(c) - \epsilon \left(\frac{u(a) - u(b)}{u(c) - u(d)} \right) u(d), \end{aligned}$$

and then you can check that (4) implies

$$\sum_{j=1}^m \lambda_j v(a_j) - \sum_{j=1}^m \mu_j v(a_j) \neq 0.$$

Thus, given λ and μ as above, u and v cannot both satisfy (1), a final contradiction that completes the proof. ■

Let's consider some implications of Theorem 2.1. First, suppose u and v are both von Neumann-Morgenstern representation with $v = \alpha u + \beta$, take any $a, b, c, d \in A$, and suppose

$$u(a) - u(b) > u(c) - u(d).$$

Then

$$(\alpha u(a) + \beta - \alpha u(b) - \beta) > (\alpha u(c) + \beta - \alpha u(d) - \beta),$$

which is equivalent to

$$v(a) - v(b) > v(c) - v(d).$$

Therefore, unlike an ordinal utility function, u does give us a way of comparing magnitudes of utility differences: if the increase in utility going from b to a is larger for u than the increase going from d to c , then it is larger for all von Neumann-Morgenstern representations. We can actually say something more: given any two pairs of alternatives, the ratio of utility differences

$$\frac{u(a) - u(b)}{u(c) - u(d)}$$

is the same for all von Neumann-Morgenstern representation! You should prove this for yourself and show that this implies the first claim.

Does this give us an unambiguous measure of how many times bigger the increase in welfare going from b to a is compared to the increase going from d to c ? Well, sort of. We just have to be careful about how we interpret the term “welfare” here. To be precise, it tells us about the decision maker’s willingness to trade off probabilities on certain outcomes. Suppose, for the sake of argument, that $u(a) > u(b)$ and $u(c) > u(d)$. If we take ϵ probability from a and put it on b , we have to move

$$\frac{u(a) - u(b)}{u(c) - u(d)}$$

times more probability from d to c to compensate the decision maker.

From a more practical perspective, Theorem 2.1 allows us to pin down the values of a von Neumann-Morgenstern representation at any two alternatives. In the construction in the proof of Theorem 1.1, we provided a u that attained a maximum value of one and a minimum value of zero; in fact, this can be done as long as the decision maker is not completely different over all lotteries. In particular, suppose we are given one von Neumann-Morgenstern representation u ; then define $\hat{u}$ by the following positive affine transformation:

$$\hat{u}(a_j) = \frac{1}{\max_{h=1,\dots,m} u(a_h) - \min_{h=1,\dots,m} u(a_h)} \left[u(a_j) - \min_{h=1,\dots,m} u(a_h) \right].$$

This gives us an equally valid representation of the decision maker’s preferences, and it attains a maximum value of one and minimum of zero.

Clearly, u is different from the notion of an ordinal utility function. A term that is also sometimes used for u to make this distinction clear is “cardinal utility function.” This jargon is used in other ways, however, so we adhere to the terminology of “expected utility representation” or “von Neumann-Morgenstern utility.”

3 States of the World

We can impose more structure on the model to analyze the decision maker’s problem. Assume there are ℓ possible *states of the world*, $s_1, \dots, s_\ell$, which summarize all details of the world that are relevant to the decision maker but unknown to her at the time she makes a choice. The decision maker’s beliefs about the state of the world are represented by a probability distribution $\pi = (\pi_1, \dots, \pi_\ell)$, where π_j is the probability of state s_j . The decision maker makes one of k choices, $c_1, \dots, c_k$. To map this model into the previous framework, an alternative is a pair $a = (c_i, s_j)$, and the set of alternatives is

$$A = \{c_1, \dots, c_k\} \times \{s_1, \dots, s_\ell\}.$$

The decision maker has an ordering on the set of lotteries over A satisfying Independence and Intermediate values, and by Theorem 1.1, we may represent those preferences succinctly by a von Neumann-Morgenstern representation u defined on A .

Each choice c_i determines the following lottery on A : alternative (c_h, s_j) is realized with probability

$$\Pr(c_h, s_j) = \begin{cases} \pi_j & \text{if } h = i, \\ 0 & \text{else.} \end{cases}$$

That is, the decision maker's choice determines the second component of $a = (s, c)$, while uncertainty about the first component (the state of the world) is summarized by the probability distribution π . Then the decision maker's expected utility from choosing c_i is

$$U(c_i|\pi) = \sum_{j=1}^{\ell} \pi_j u(c_i, s_j),$$

where we modify notation slightly to bring out the dependence of U on the choice and the beliefs about the state of the world. When π is degenerate on a single state s_j , we write the expected utility as $U(c_i|s_j)$.

The decision maker's problem is to make the choice that maximizes her expected utility given her beliefs about the state of the world:

$$\max_{i=1, \dots, k} U(c_i|\pi).$$

Clearly, there will always be a choice c_i that solves this problem. There could be two optimal choices, say c_i and c_h , and then it does not matter which the decision maker makes; in fact, she could decide between them randomly without affecting her expected utility.

There may be some choices that are optimal for no beliefs; that is, some c_i may be such that for every π , there exists c_h with $U(c_h|\pi) > U(c_i|\pi)$. The statement of this condition allows c_h to vary with π ; for different beliefs, it may be that c_i is inferior to different choices. This is immaterial, for it is still the case that c_i would never be optimal. We say c_h *strictly dominates* c_i if c_h provides a strictly higher utility than c_i in every state of the world: for all s_j , $u(c_h, s_j) > u(c_i, s_j)$. Then it is obvious that c_i is never optimal, and it is actually the case that a single choice is superior to c_i in every state of the world. Thus, if c_i is strictly dominated by some other choice, an implication is that it is never optimal for any beliefs, but strict dominance is much stronger than sub-optimality.

To give a complete characterization of sub-optimality, we must consider the possibility that the decision maker randomizes over choices. A *mixed choice*, denoted $\gamma =$

$(\gamma_1, \dots, \gamma_k)$, is a probability distribution over choices $c_1, \dots, c_k$. The interpretation is that c_i is chosen with probability γ_i . We view each “pure” choice c_i as a special case of mixed choice γ in which γ is degenerate on c_i . The expected utility from a mixed choice γ is

$$U(\gamma|\pi) = \sum_{i=1}^k \gamma_i U(c_i|\pi).$$

We extend the notion of strict dominance in the straightforward way: say γ *strictly dominates* c_i if for all s_j , $U(\gamma|s_j) > U(c_i|s_j)$. We have already noted that there will always be a choice that maximizes the decision maker’s expected utility, and there is no way to increase the expected utility of the optimal choice by mixing. Still, it may be that a pure choice is inferior to a mixed choice. Perhaps surprisingly, it may be that a pure choice is not strictly dominated by any other pure choice, but it is strictly dominated by a *mixed* choice!

In the following example, there are three states, and none of the three possible choices is strictly dominated by any other. Note, however, that the mixed choice $\gamma = (\frac{1}{2}, \frac{1}{2}, 0)$ that places equal probability on choices c_1 and c_2 generates an expected utility of two in each of the two states. Therefore, γ strictly dominates c_3 .

	s_1	s_2
c_1	4	0
c_2	0	4
c_3	1	1

Note that if c_i is strictly dominated by a mixed choice γ , then it is never optimal. Indeed, consider any beliefs π . By strict dominance, we have $U(\gamma|s_j) > U(c_i|s_j)$ for all states, and therefore $U(\gamma|\pi) > U(c_i|\pi)$. Rewriting this, we have

$$\sum_{j=1}^{\ell} \gamma_j U(c_j|\pi) > U(c_i|\pi), \quad (5)$$

so at least one term on the left is strictly greater than $U(c_i|\pi)$, i.e., $U(c_j|\pi) > U(c_i|\pi)$ for some c_j , so c_i is not optimal.

The next result provides a complete characterization of sub-optimality, showing that the opposite direction of the above claim holds: a choice is optimal for no beliefs if and only if it is strictly dominated by some mixed choice. Interestingly, the proof is geometric, relying on the separating hyperplane theorem.

Theorem 3.1. *The following conditions on a choice c_i are equivalent:*

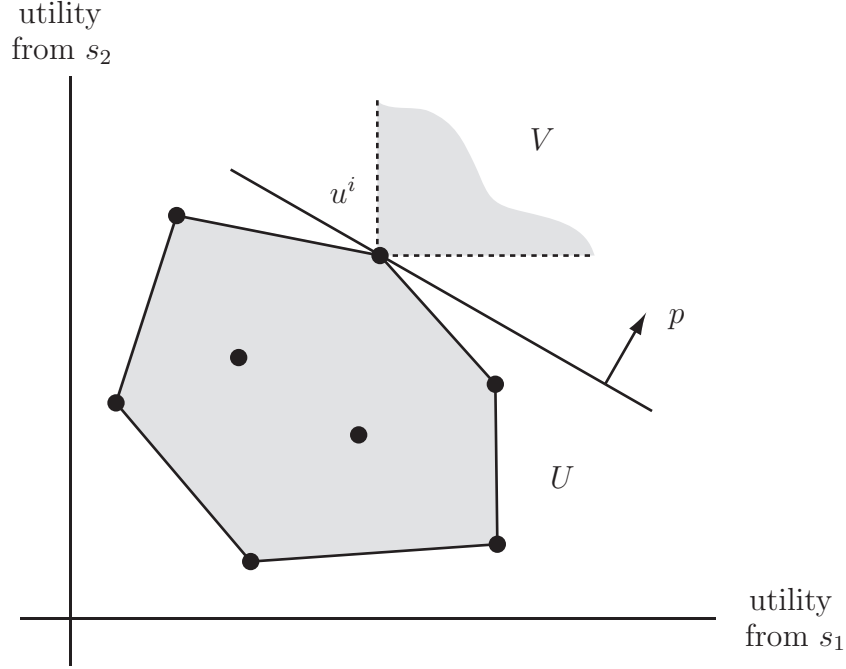


Figure 4: Separating hyperplane

- (a) for every belief π , there exists c_j such that $U(c_j|\pi) > U(c_i|\pi)$,
- (b) there is a mixed choice γ that strictly dominates c_i .

Proof. We have already argued that (b) implies (a). Assume (a) holds. To every choice c_h , associate a vector $u^h = (u(c_h|s_1), \dots, u(c_h|s_\ell)) \in \mathbb{R}^\ell$ of utilities corresponding to the different states of the world. Note that for every π , $U(c_h|\pi) = \pi \cdot u^h$. Furthermore, the vector of utilities associated with the mixed choice γ is the convex combination $\sum_{h=1}^k \gamma_h u^h$. Let

$$U = \text{conv}\{u^1, \dots, u^k\}$$

denote the vectors of utilities generated by all mixed choices. Let

$$V = \{v \in \mathbb{R}^m \mid v \gg u^i\}$$

denote the set of vectors that are strictly greater than u^i in each coordinate. See Figure 4 for the case of two states. Suppose in order to deduce a contradiction that there is no mixed choice that strictly dominates c_i ; in other words, $U \cap V = \emptyset$. Both sets U and V are convex. By the separating hyperplane theorem, there exists $p \in \mathbb{R}^m \setminus \{0\}$ that weakly separates U and V , i.e., for all $u \in U$ and all $v \in V$, we have

$$p \cdot u \leq p \cdot u^i \leq p \cdot v,$$

as in the figure. I claim that $p \geq 0$, i.e., $p_j \geq 0$ for all $j = 1, \dots, \ell$. Suppose, instead, that $p_j < 0$ for some coordinate. But then

$$u(\alpha) = (u_1^i, \dots, u_{j-1}^i, u_j^i + \alpha, u_{j+1}^i, \dots, u_\ell^i) \in V,$$

and we could choose α sufficiently large that $p \cdot v < p \cdot u^i$. Therefore, $p \geq 0$, and $\sum_{j=1}^{\ell} p_j > 0$, so we can define

$$\pi = \frac{1}{\sum_{j=1}^{\ell} p_j} p. \quad (6)$$

Then the inequality $\pi \cdot u \leq \pi \cdot u^i$ continue to hold for all $u \in U$. But note that π is a probability distribution over states, and c_i maximizes the decision maker's expected utility over all choices, contradicting (a). $\blacksquare$

We can extend the analysis: say γ *weakly dominates* c_i if $U(\gamma|s_j) \geq U(c_i|s_j)$ for all j , with strict inequality for at least one state. A belief π is *totally mixed* if it places positive probability on each state, i.e., $\pi_j > 0$ for all j . The next result establishes that a choice is optimal for some totally mixed belief if and only if it is not weakly dominated by a mixed choice.

Theorem 3.2. *The following conditions on a choice c_i are equivalent:*

- (a) *for every totally mixed belief π , there exists c_j such that $U(c_j|\pi) > U(c_i|\pi)$,*
- (b) *there is a mixed choice γ that weakly dominates c_i .*

The proof that (b) implies (a) is similar to the above, now using the assumption that each π_j is positive to obtain the strict inequality in (5). The proof that (a) implies (b) follows the proof for Theorem 3.1; we need only modify the definition of V as the set of vectors that weakly dominate u^i , and we show that $p \gg 0$. Details are omitted.

4 The One-dimensional Choice Model

In the one-dimensional choice model, where $A \subseteq \mathbb{R}$, the set of alternatives may be infinite, and uncertainty may best be modelled by a continuous distribution, e.g., the normal distribution when $A = \mathbb{R}$. The axiomatic result of Theorem 1.1 carries over to this more general setting, though now expected utility may be calculated as the integral of the decision maker's von Neumann-Morgenstern representation with respect to a density. Although a lottery λ to be continuously distributed, we continue to allow lotteries that are represented by a discrete distribution with finite support. In this case, letting $\lambda(x)$ denote the probability of x under lottery λ , there exist a natural number m and alternatives $x_1, x_2, \dots, x_m \in A$ such that $\sum_{j=1}^m \lambda(x_j) = 1$. Such lotteries are called *simple*.

Because the outcomes of the lottery are now real numbers, we can define the probability, $F(x)$, that the realization from λ is less than or equal to x . Formally, F is the *distribution* of λ . When λ is simple, it is defined as

$$F(x) = \sum_{j: x_j \leq x} \lambda(x_j)$$

for all $x \in \mathbb{R}$. When λ is continuously distributed, the distribution F may be any continuous, weakly increasing function satisfying $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow \infty} F(x) = 1$; and in this case, we denote by f the density of F , so $F(b) - F(a) = \int_a^b f(x)dx$ for all $a, b \in \mathbb{R}$. In addition, we can now define the concept of the expected value, or *mean*, of a lottery. When λ is simple, this is

$$E[\lambda] = \sum_{j=1}^m x_j \lambda(x_j),$$

and when λ is continuously distributed, it is

$$E[\lambda] = \int_{-\infty}^{\infty} x f(x) dx.$$

We usually distinguish the distribution of a lottery μ , distinct from λ , with the notation G , distinct from F .

As an example of a choice problem that involves lotteries over monetary outcomes, consider a consumer's decision about whether to purchase a lottery ticket with the following properties: the ticket costs one dollar and pays one million dollars if it wins; suppose the lottery is "fair," i.e., the chance of winning is one in a million. Let the outcome x of the lottery denote the change in the consumer's income. Then the lottery ticket yields $x = -1$ with probability $999,999/1,000,000$ and $x = 999,999$ with probability $1/1,000,000$. Not buying the lottery ticket yields outcome $x = 0$ with probability one. Should the consumer purchase the lottery ticket? That depends on which lottery he or she prefers. Suppose the consumer's von Neumann-Morgenstern representation were $u(x) = x$. You can check that the expected utility of buying the lottery ticket would be zero (in fact, that is the expected value of the lottery), which is the same as the expected utility of not buying. Thus, the consumer would be indifferent. If the probability of winning the lottery were higher or lower, the consumer would have a corresponding strict preference for buying or not buying the ticket, respectively.

Since most lotteries are not fair, does this mean that no one should buy lottery tickets? No, it depends on the consumer's von Neumann-Morgenstern representation. If

$$u(x) = \begin{cases} x & \text{if } x < 0 \\ 2x & \text{if } x \geq 0, \end{cases}$$

then the consumer would have a strictly higher expected utility,

$$\left(\frac{999,999}{1,000,000}\right)(-1) + \left(\frac{1}{1,000,000}\right)(1,999,998) = \frac{999,999}{1,000,000},$$

from buying the lottery ticket. If

$$u(x) = \begin{cases} 2x & \text{if } x < 0 \\ x & \text{if } x \geq 0, \end{cases}$$

then the consumer would have a strictly lower expected utility,

$$\left(\frac{999,999}{1,000,000}\right)(-2) + \left(\frac{1}{1,000,000}\right)(1,000,000) = -\frac{999,999}{1,000,000},$$

from buying the lottery ticket. Thus, the consumer's decision depends on his or her "risk preferences."

Letting $\delta_{E[\lambda]}$ denote the degenerate lottery with probability one on the mean of λ , we say the decision maker is *risk averse* if for all non-degenerate lotteries λ on A , we have $\delta_{E[\lambda]} P \lambda$. We say the decision maker is *risk loving* if for all non-degenerate λ , we have $\lambda P \delta_{E[\lambda]}$. And we say the decision maker is *risk neutral* if for all λ , we have $\delta_{E[\lambda]} I \lambda$.

The next result characterizes the decision maker's attitude toward risk in terms of the curvature of the von Neumann-Morgenstern representation.

Theorem 4.1. *Assume $A \subseteq \mathbb{R}$ is convex, and let u be any von Neumann-Morgenstern representation.*

1. *A decision maker is risk averse if and only if u is strictly concave.*
2. *A decision maker is risk loving if and only if u is strictly convex.*
3. *A decision maker is risk neutral if and only if u is affine linear.*

In fact, part 1 of Theorem 4.1 is a version of "Jensen's inequality," which states that if a function u is concave, then the expected value of the function with respect to a lottery λ is less than or equal to the function evaluated at the mean of λ :

$$u(E[\lambda]) \geq \sum_{j=1}^m \lambda(x_j) u(x_j) \quad \text{or} \quad u(E[\lambda]) \geq \int u(x) f(x) dx,$$

depending on whether λ is simple or continuous. This in turn follows from the separating hyperplane theorem, as depicted in Figure 5. The *epigraph* of u is the subset

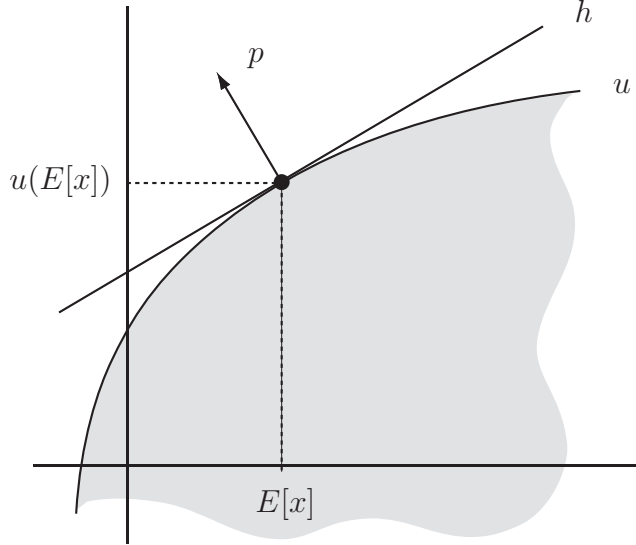


Figure 5: Jensen's Inequality

$\{(x, y) \in \mathbb{R}^2 \mid y \leq u(x)\}$ of the Cartesian plane below the graph of u ; since u is concave, its epigraph is convex, and the ordered pair $(E[\lambda], u(E[\lambda]))$ lies on the boundary of this set. By the separating hyperplane theorem, there is a linear function ϕ that weakly separates $(E[\lambda], u(E[\lambda]))$ from this set. Let ϕ have gradient $p = (p_1, p_2)$. Thus, given (x, y) such that $y \leq u(x)$,

$$p_1 x + p_2 y \leq p_1 E[\lambda] + p_2 u(E[\lambda]).$$

Furthermore, $p_2 \neq 0$,¹ and it follows that the level set of ϕ through $(E[\lambda], u(E[\lambda]))$ is the graph of an affine linear function $h: \mathbb{R} \rightarrow \mathbb{R}$; specifically,

$$h(x) = \frac{p_1 E[\lambda] + p_2 u(E[\lambda])}{p_2} + \frac{p_1}{p_2} x.$$

Then, using linearity of h and the fact that $u \leq h$, we have

$$u(E[\lambda]) = h(E[\lambda]) = h\left(\sum_{j=1}^m \lambda(x_j) x_j\right) = \sum_{j=1}^m \lambda(x_j) h(x_j) \geq \sum_{j=1}^m \lambda(x_j) u(x_j),$$

delivering Jensen's inequality. When u is strictly concave, the vector $(E[\lambda], u(E[\lambda]))$ uniquely maximizes ϕ over the epigraph of u , so if λ is non-degenerate, we obtain a strict inequality in the above displayed expression.

The usual assumption in applications of expected utility to the one-dimensional model is that of risk aversion. This is largely due to introspection: for most of us, “a bird

¹Technical point: it may actually be that $p_2 = 0$, but because u is concave, this is only possible if $E[\lambda]$ is an extreme point of A , so λ must then be degenerate, and Jensen's inequality follows trivially.

in the hand is worth two in the bush.” To see why risk neutrality, for example, is not always compelling, suppose $A = \mathbb{R}_+$ is money, suppose a decision maker has a von Neumann-Morgenstern representation $u(x) = x$, and consider the following lottery: we flip a coin until it comes up heads, and, if that takes t tosses, then we give the decision maker $x = 2^t$ dollars.²

	H	TH	TTH	TTTH	...
prize	2	4	8	16	...
prob	1/2	1/4	1/8	1/16	...

What is the most the decision maker would be willing to pay for this lottery? In other words, what is the certainty equivalent of this lottery? Because the decision maker is risk neutral, it is equal to $E[\lambda]$, which is infinite! Since most of us wouldn’t be willing to pay an arbitrarily high amount for this lottery, we’re not risk neutral or risk loving. Though it’s not really a paradox, this example is called the *St. Petersburg Paradox*. In the sequel, we explore the one-dimensional model in more detail.

5 Stochastic Dominance

In the one-dimensional choice model, we are able to make some comparisons between lotteries based on minimal structure of the decision maker’s preferences. For example, under the reasonable assumption that the decision maker prefers more money to less, so u is increasing, the lottery with probability 999,999/1,000,000 on -1 and probability 1/1,000,000 on 999,999 is less desirable than the even chance lottery on -1 and 999,999. The difference between these lotteries is that we have moved probability from worse outcomes to better ones, which must be an improvement for the decision maker. Formally, we say λ *first order stochastically dominates* μ if for all $x \in \mathbb{R}$, $F(x) \leq G(x)$, where F is the distribution of λ and G is the distribution of μ . Note the latter inequality may be the reverse of that expected; think of probability being “squeezed” to the right up the real line. See Figure 6 for continuous and discrete pictures of a lottery with distribution F that stochastically dominates a lottery with distribution G in the first order sense.

In fact, first order stochastic dominance is characterized by the property that it always amounts to an improvement for a decision maker with monotonic preferences.

Theorem 5.1. *Assume $A = \mathbb{R}$, and let λ and μ be any two lotteries on $\mathbb{R}$ with bounded support. Then λ first order stochastically dominates μ if and only if for every weakly increasing von Neumann-Morgenstern utility representation u , we have $U(\lambda) \geq U(\mu)$.*

²Admittedly, this is not a simple lottery, because it puts positive probability on an infinite number of outcomes, but it makes a nice point.

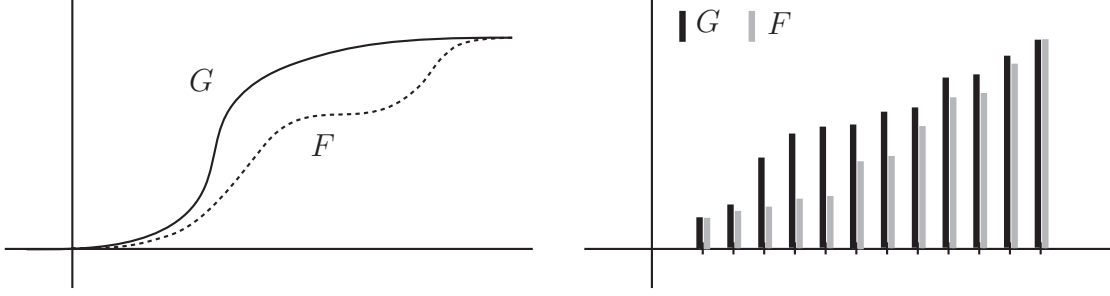


Figure 6: First order stochastic dominance

Proof. First, assume λ first order stochastically dominates μ , and let u be weakly increasing. For simplicity, let λ have distribution F and density f , while μ has distribution G and density g , and let u be differentiable. We need to show $\int_{-\infty}^{\infty} u(x)f(x)dx \geq \int_{-\infty}^{\infty} u(x)g(x)dx$, or equivalently,

$$\int_{-\infty}^{\infty} u(x)(f(x) - g(x))dx \geq 0.$$

Using integration by parts, this is equivalent to

$$\lim_{x \rightarrow \infty} u(x)(F(x) - G(x)) - \lim_{x \rightarrow -\infty} u(x)(F(x) - G(x)) \geq \int_{-\infty}^{\infty} u'(x)(F(x) - G(x))dx,$$

and because the lotteries have bounded support, the lefthand side is just zero. And for each x , $u'(x) \geq 0$ and $F(x) - G(x) \leq 0$, so the lefthand side is non-positive, as required. Now assume that for all weakly increasing functions u , we have $\int_{-\infty}^{\infty} u(x)f(x)dx \geq \int_{-\infty}^{\infty} u(x)g(x)dx$. Consider any $x \in \mathbb{R}$. To show $F(x) \leq G(x)$, define u as

$$u(y) = \begin{cases} 0 & \text{if } y \leq x \\ 1 & \text{else.} \end{cases}$$

Then

$$1 - F(x) = \int_{-\infty}^{\infty} u(y)f(y)dy \geq \int_{-\infty}^{\infty} u(y)g(y)dy = 1 - G(x),$$

which implies $F(x) \leq G(x)$, completing the proof. ■

Thus, in general, if u is weakly increasing, then a first order stochastic increase in λ leads to a higher expected value of u . This remains true if u is also concave. But with the added structure of concavity, it may be that the expected value of u is increased by a larger class of transformations of λ . Returning to the lottery example, suppose that a lottery ticket costs only 50 cents and pays off 500,000 dollars, i.e., the lottery ticket yields $x = -.5$ with probability 999,999/1,000,000 and $x = 499,999.5$ with probability 1/1,000,000. The expected payoff of this lottery is still zero, but

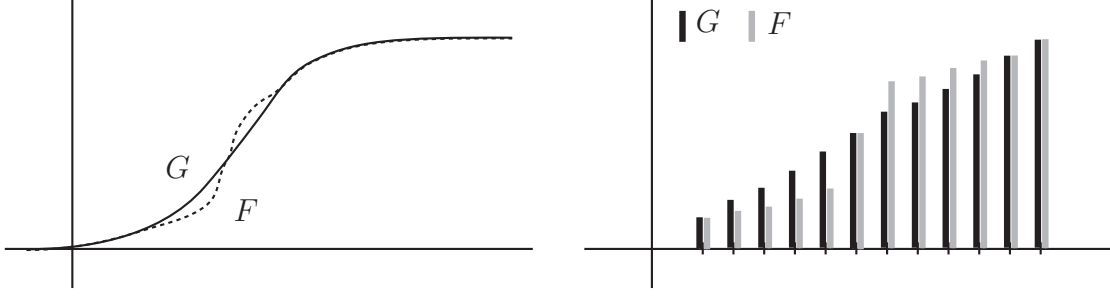


Figure 7: Second order stochastic dominance

compared to the original example, the downside and the upside of the lottery are reduced by half; the mean of the lottery is unchanged, but the risk is reduced. This should be preferable for a risk averse decision maker.

In fact, this transformation is an example of another type of stochastic dominance. We say λ *second order stochastically dominates* μ if for all $x \in \mathbb{R}$, $\int_{-\infty}^x F(y)dy \leq \int_{-\infty}^x G(y)dy$, where F and G are, respectively, the distributions of λ and μ . Obviously, first order stochastic dominance implies second order stochastic dominance, i.e., there are more lotteries that dominate λ in the second order sense than in the first order sense. See Figure 7 for continuous and discrete pictures of second order stochastic dominance. In fact, second order stochastic dominance is characterized by the property that it always increases the expected utility of a risk averse decision maker with monotonic preferences.

Theorem 5.2. *Assume $A = \mathbb{R}$, and let λ and μ be any two lotteries on $\mathbb{R}$ with bounded support. Then λ second order stochastically dominates μ if and only if for every weakly increasing and concave von Neumann-Morgenstern utility representation u , we have $U(\lambda) \geq U(\mu)$.*

Proof. First, assume λ second order stochastically dominates μ , and let u be weakly increasing and concave. For simplicity, let λ have distribution F and density f , while μ has distribution G and density g , and let u be twice differentiable. As in the proof of Theorem 5.1, we need to show

$$0 \geq \int_{-\infty}^{\infty} u'(x)(F(x) - G(x))dx.$$

Using integration by parts again, the righthand side is equal to

$$\begin{aligned} & \lim_{x \rightarrow \infty} [u'(x) \int_{-\infty}^x (F(y) - G(y))dy] - \lim_{x \rightarrow -\infty} [u'(x) \int_{-\infty}^x (F(y) - G(y))dy] \\ & - \int_{-\infty}^{\infty} u''(x) \int_{-\infty}^x (F(y) - G(y))dy dx. \end{aligned}$$

Since the lotteries have bounded support, the first two terms above are zero. And for each x , $u''(x) \leq 0$ and $\int_{-\infty}^x (F(y) - G(y))dy \leq 0$, so the entire expression is non-positive, as required. Now assume that for all weakly increasing and concave functions u , we have $\int_{-\infty}^{\infty} u(x)f(x)dx \geq \int_{-\infty}^{\infty} u(x)g(x)dx$. Consider any $x \in \mathbb{R}$. To show $\int_{-\infty}^x F(y)dy \leq \int_{-\infty}^x G(y)dy$, define the weakly increasing and concave function u as

$$u(y) = \begin{cases} y & \text{if } y \leq x, \\ x & \text{else.} \end{cases}$$

Then using integration by parts twice, we have

$$\begin{aligned} x - \int_{-\infty}^x F(y)dy &= xF(x) - \lim_{y \rightarrow -\infty} yF(y) - \int_{-\infty}^x F(y)dy + x(1 - F(x)) \\ &= \int_{-\infty}^x yf(y)dy + \int_x^{\infty} xf(y)dy \\ &= \int_{-\infty}^x u(y)f(y)dy \\ &\geq \int_{-\infty}^x u(y)g(y)dy \\ &= \int_{-\infty}^x yg(y)dy + \int_x^{\infty} xg(y)dy \\ &= xG(x) - \lim_{y \rightarrow -\infty} yG(y) - \int_{-\infty}^x G(y)dy + x(1 - G(x)) \\ &= x - \int_{-\infty}^x G(y)dy, \end{aligned}$$

which implies $\int_{-\infty}^x F(y)dy \leq \int_{-\infty}^x G(y)dy$, completing the proof. ■

6 Measures of Risk Aversion

It is typical to assume $A \subseteq \mathbb{R}$ is a convex subset of the real line and a continuous von Neumann-Morgenstern representation u . Then, by the intermediate value theorem, for every lottery λ on A there is an alternative $x \in A$ such that $u(x) = U(\lambda)$. In fact, we will assume that u is twice continuously differentiable. In many applications, alternatives represent a good (such as money) for the decision maker, so we now assume that $A = \mathbb{R}_+$ and u is strictly increasing as well. Then there is exactly one $x \in A$ such that $u(x) = U(\lambda)$. We call this the (*absolute*) *certainty equivalent* of λ , and we denote it $C^a[\lambda]$. If the decision maker is risk averse, it follows that $C^a[\lambda] \leq E[\lambda]$. (Why?) We refer to the difference $R^a[\lambda] = E[\lambda] - C^a[\lambda]$ as the (*absolute*) *risk premium* of λ . Obviously, if the decision maker is risk neutral, then $C^a[\lambda] = E[\lambda]$ and the risk premium is zero. See Figure 8.

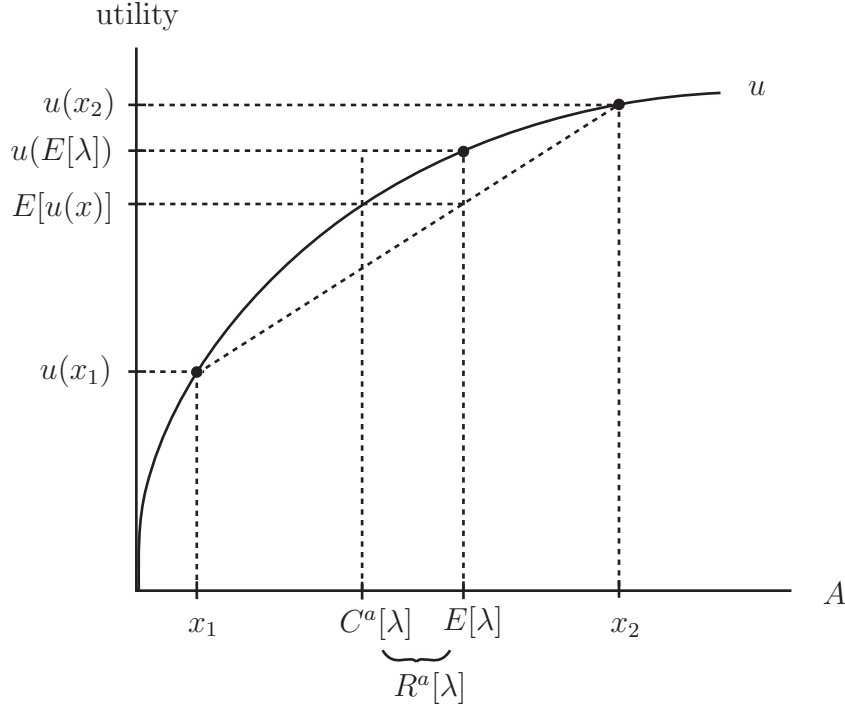


Figure 8: Risk aversion

We can define relative versions of these quantities, as well. Given any lottery λ , define the *relative certainty equivalent* of λ , denoted $C^r[\lambda]$, implicitly by $u(C^r[\lambda]E[\lambda]) = U(\lambda)$. If the decision maker is risk averse, it follows that $C^r[\lambda] \leq 1$. (Right?) Then the fraction $R^r[\lambda] = 1 - C^r[\lambda]$ is the *relative risk premium* of λ . Obviously, if the decision maker is risk neutral, then $C^r[\lambda] = 1$ and the relative risk premium is zero.

So far, we have the necessary conceptual apparatus to categorize a decision maker as risk averse or not, but we have not attempted to measure the level of risk aversion. One approach would be to compare the risk aversion of two decision makers by comparing their respective certainty equivalents. But certainty equivalents are defined with respect to a given lottery, so this would give us a measure of risk aversion with respect to some λ . To develop a measure that is independent of any particular lottery, we consider risk preferences over “local” lotteries at a given initial monetary level. Formally, our measure of the *absolute risk aversion* of u at x is

$$-\frac{u''(x)}{u'(x)}.$$

This provides a local measure of risk aversion at a monetary level x . It may be that u is more risk averse than some other $\tilde{u}$ around x but less so around y . Note that risk aversion essentially implies $u''(x) < 0$, so the above measure is positive, with higher measures indicating less acceptance of risk. Note also that the measure of absolute

risk aversion is invariant with respect to affine transformations of u :

$$-\frac{D^2(\alpha u(x) + \beta)}{D(\alpha u(x) + \beta)} = -\frac{\alpha u''(x)}{\alpha u'(x)} = -\frac{u''(x)}{u'(x)}.$$

Thus, it depends only on the decision maker's risk preferences, and not on any cardinal properties of a particular von Neumann-Morgenstern representation.

Interestingly, the absolute risk aversion of u at x is related to the risk premium of small, fair lotteries around x . Given $\epsilon > 0$, let $\lambda(\epsilon)$ be the even chance lottery between $x + \epsilon$ and $x - \epsilon$. To simplify notation, let $C^a(\epsilon)$ denote the certainty equivalent of $\lambda(\epsilon)$, and let $R^a(\epsilon)$ denote the corresponding risk premium. Then, by definition, $C^a(\epsilon)$ satisfies

$$u(C^a(\epsilon)) = \frac{1}{2}u(x - \epsilon) + \frac{1}{2}u(x + \epsilon)$$

for all $\epsilon \in (0, x)$. Differentiating this identity with respect to ϵ , we have

$$u'(C^a(\epsilon)) \frac{d}{d\epsilon} C^a(\epsilon) = -\frac{1}{2}u'(x - \epsilon) + \frac{1}{2}u'(x + \epsilon),$$

and therefore, $\frac{d}{d\epsilon} C^a(0) = 0$. With this in mind, we take the second derivative of the above identity to arrive at

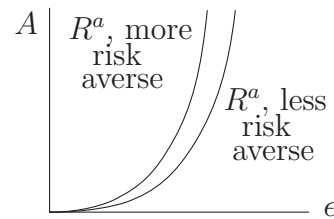
$$u''(C^a(\epsilon)) \left[\frac{d}{d\epsilon} C^a(\epsilon) \right]^2 + u'(C^a(\epsilon)) \frac{d^2}{d\epsilon^2} C^a(\epsilon) = \frac{1}{2}u''(x - \epsilon) + \frac{1}{2}u''(x + \epsilon)$$

Evaluating this at $\epsilon = 0$ and substituting into $\frac{d^2}{d\epsilon^2} R^a(0) = -\frac{d^2}{d\epsilon^2} C^a(0)$, it follows that

$$\frac{d^2}{d\epsilon^2} R^a(0) = -\frac{u''(x)}{u'(x)},$$

so the absolute risk aversion of u at x is just the second derivative of the risk premium for small gambles around x . This reveals the behavioral content of this measure: if the absolute risk aversion of u at x is greater than that of $\tilde{u}$, then a decision maker whose preferences are characterized by u is willing to pay a higher premium to avoid the risk of small gambles around x .

To elaborate, if the absolute risk aversion of u at x exceeds that of $\tilde{u}$, then the first derivative of $R^a(\epsilon)$ at zero is increasing faster for u than it is for $\tilde{u}$; by twice continuous differentiability, this is true for ϵ close to zero. Since the first derivative $\frac{d}{d\epsilon} R^a(0)$ is equal to zero for both utilities, it follows that the risk premium of u exceeds that of $\tilde{u}$ for small ϵ , as depicted to the right.



When u is strictly concave but not increasing, i.e., it achieves a maximum on A , we can extend the analysis. Now the certainty equivalent is not uniquely defined, as there are two solutions to the equation in the definition of that concept. That is, there are $C^{a-}[\lambda]$ and $C^{a+}[\lambda]$ such that $C^{a-}[\lambda] \leq E[\lambda] \leq C^{a+}[\lambda]$ and

$$u(C^{a-}[\lambda]) = u(C^{a+}[\lambda]) = U(\lambda),$$

and two corresponding definitions of risk premium,

$$\begin{aligned} R^{a-}[\lambda] &= E[\lambda] - C^{a-}[\lambda] \\ R^{a+}[\lambda] &= C^{a+}[\lambda] - E[\lambda], \end{aligned}$$

both defined to be positive numbers indicating how far (to the left or right) from the mean of λ the decision maker would be willing to move to avoid risk. Following the above, we can define the risk premiums $R^{a-}(\epsilon)$ and $R^{a+}(\epsilon)$ corresponding to small, fair gambles around a given x . Although for a given positive ϵ , these premiums may differ, the derivative of each at $\epsilon = 0$ is zero, and the second derivatives are equal:

$$\frac{d}{d\epsilon} R^{a-}(0) = \frac{d}{d\epsilon} R^{a+}(0) = 0 \quad \text{and} \quad \frac{d^2}{d\epsilon^2} R^{a-}(0) = \frac{d^2}{d\epsilon^2} R^{a+}(0).$$

Again, the absolute risk aversion of u at x is the second derivative of R^{a-} and R^{a+} at $\epsilon = 0$, with the same interpretation as before.

Returning to the assumption that u is strictly increasing, we can define *relative risk aversion* of u at x ,

$$-\frac{xu''(x)}{u'(x)},$$

in terms of relative certainty equivalents. Given $\epsilon > 0$, let $\lambda(\epsilon)$ be the even chance lottery between $(1 + \epsilon)x$ and $(1 - \epsilon)x$. Let $C^r(\epsilon)$ be the relative certainty equivalent of this lottery and $R^r(\epsilon)$ the relative risk premium. By definition,

$$u(C^r(\epsilon)x) = \frac{1}{2}u((1 - \epsilon)x) + \frac{1}{2}u((1 + \epsilon)x),$$

and differentiating with respect to ϵ ,

$$u'(C^r(\epsilon)x) \frac{d}{d\epsilon} C^r(\epsilon)x = -\frac{x}{2}u'((1 - \epsilon)x) + \frac{x}{2}u'((1 + \epsilon)x),$$

which implies $\frac{d}{d\epsilon} C^r(0) = 0$. Taking the second derivative, we have

$$u''(C^r(\epsilon)x) \left[\frac{d}{d\epsilon} C^r(\epsilon)x \right]^2 + u'(C^r(\epsilon)x) \frac{d^2}{d\epsilon^2} C^r(\epsilon)x = \frac{x^2}{2}u''((1 - \epsilon)x) + \frac{x^2}{2}u''((1 + \epsilon)x).$$

Evaluating this at $\epsilon = 0$ and substituting into $R^r(0) = 1 - C^r(0)$, it follows that

$$\frac{d^2}{d\epsilon^2} R^r(0) = -\frac{xu''(x)}{u'(x)},$$

so the relative risk aversion of u at x is just the second derivative of the relative risk premium for small gambles (in percentage terms) around x . Again, the analysis extends to the general case of strictly concave, but possibly single-peaked, von Neumann-Morgenstern representations.

7 The Multidimensional Choice Model

Everything carries over. We define simple lotteries and risk aversion (loving, neutrality) just as in the one-dimensional choice model, and Theorems 1.1, 2.1, and 4.1 hold in their original forms. Furthermore, everything carries over to lotteries on $\mathbb{R}^n$ with continuous distributions too. Of course, the analysis of risk premiums and stochastic dominance relies on the structure of the one-dimensional choice model.

8 Mean-Variance Analysis

In the multidimensional choice model, it turns out that quadratic von Neumann-Morgenstern representations are especially easy to work with. The decision maker's expected utility from a lottery can be decomposed into two terms, one corresponding to the mean of the lottery and one corresponding to the variance. Recall that the mean and variance of a lottery λ are defined as

$$\begin{aligned} E[\lambda] &= \sum_{j=1}^m \lambda(x_j) x_j \\ V[\lambda] &= \sum_{j=1}^m \lambda(x_j) \|x_j - E[\lambda]\|^2 \end{aligned}$$

when λ is simple. Of course, the above expressions are defined analogously in terms of density when λ has a continuous distribution.

Theorem 8.1. *Assume $A \subseteq \mathbb{R}$, and the von Neumann-Morgenstern representation u is quadratic with ideal point $\hat{x}$, i.e., $u(x) = -\|x - \hat{x}\|^2$. Then for every lottery λ , we have*

$$U(\lambda) = u(E[\lambda]) - V[\lambda].$$

Proof. For simplicity, we provide the argument for the case of a simple lottery. Note

that

$$\begin{aligned}
U(\lambda) &= \sum_{j=1}^m \lambda(x_j) u(x_j) \\
&= \sum_{j=1}^m -\lambda(x_j) (x - \hat{x}) \cdot (x - \hat{x}) \\
&= - \sum_{j=1}^m \lambda(x_j) [x_j \cdot x_j - 2x \cdot \hat{x} + \hat{x} \cdot \hat{x}] \\
&= - \sum_{j=1}^m \lambda(x_j) \left[E[\lambda] \cdot E[\lambda] - 2x_j \cdot \hat{x} + \hat{x} \cdot \hat{x} \right. \\
&\quad \left. + x_j \cdot x_j - 2x_j \cdot E[\lambda] + 2x_j \cdot E[\lambda] - E[\lambda] \cdot E[\lambda] \right] \\
&= - \sum_{j=1}^m \lambda(x_j) [E[\lambda] \cdot E[\lambda] - 2x_j \cdot \hat{x} + \hat{x} \cdot \hat{x}] \\
&\quad + \sum_{j=1}^m \lambda(x_j) [x_j \cdot x_j - 2x_j \cdot E[\lambda]] + E[\lambda] \cdot E[\lambda] \\
&= E[\lambda] \cdot E[\lambda] - 2E[\lambda] \cdot \hat{x} + \hat{x} \cdot \hat{x} \\
&\quad - \sum_{j=1}^m \lambda(x_j) [x_j \cdot x_j - 2x_j \cdot E[\lambda] + E[\lambda] \cdot E[\lambda]] \\
&= -(E[\lambda] - \hat{x}) \cdot (E[\lambda] - \hat{x}) \\
&\quad - \sum_{j=1}^m \lambda(x_j) [(x_j - E[\lambda]) \cdot (x_j - E[\lambda])] \\
&= u(E[\lambda]) - V[\lambda],
\end{aligned}$$

where the first equality follows by definition of expected utility, the second from definition of quadratic utility, the third from expansion, the fourth from adding and subtracting like terms, the fifth from breaking the sum, the sixth because $E[\lambda]$ and $\hat{x}$ are constants independent of j , the seventh from gathering terms, and the last from definition of quadratic utility, mean, and variance. $\blacksquare$

Theorem 8.1 has interesting implications for the one-dimensional model with quadratic utilities. Assume there is an odd number n of voters with ideal points indexed $\hat{x}^1 < \hat{x}^2 < \dots < \hat{x}^n$, and let $m = \frac{n+1}{2}$ be the index of the voter with the median ideal point. It follows that a majority of voters prefer one lottery λ to another μ if and only if the median voter prefers λ to μ . Indeed, suppose the median voter's expected utility is such that $U_m(\lambda) > U_m(\mu)$. Without loss of generality, suppose

$E[\lambda] \leq E[\mu]$, and consider any $i > m$. Then

$$u_i(E[\lambda]) - u_i(E[\mu]) + (V[\mu] - V[\lambda]) \geq u_i(E[\lambda]) - u_i(E[\mu]) + (V[\mu] - V[\lambda]) > 0,$$

and we conclude that $U_i(\lambda) > U_i(\mu)$, as claimed.

9 Experimental Evidence

We can empirically test the accuracy of the Independence and Intermediate values axioms. In fact, there has been much work in experimental psychology and economics to see whether these axioms are consistent with observed choice behavior. Though they often are, the empiricists have noted some choice situations that are especially problematic for expected utility theory.

One such situation is as follows. Suppose there are three possible wealth levels,

$$\begin{aligned} a_1 &= 0 \\ a_2 &= 1,000,000 \\ a_3 &= 5,000,000, \end{aligned}$$

and four possible lotteries over them,

- (a) $(0, 1, 0)$
- (b) $(.01, .89, .10)$
- (c) $(.89, .11, 0)$
- (d) $(.90, 0, .10)$.

Experiments have shown that subjects tend to prefer lottery (a) to (b) and lottery (d) to (c).

If such a decision maker has an ordering of lotteries satisfying Independence and Intermediate values, then there is a von Neumann-Morgenstern representation, say u , representing those preferences. A preference for (a) over (b) implies

$$u(a_2) > (.01)u(a_1) + (.89)u(a_2) + (.10)u(a_3),$$

or equivalently,

$$(.11)u(a_2) > (.01)u(a_1) + (.10)u(a_3).$$

But a preference for (d) over (c) implies the *opposite* inequality, an impossibility. Therefore, the decision maker's preferences over lotteries cannot satisfy the von Neumann-Morgenstern axioms: either Independence or Intermediate values is violated. This is known as the *Allais Paradox*.

10 The Paradox of Voting

As a simple application of expected utility theory, consider the problem of a voter who is considering whether to vote for candidate A , to vote for candidate B , or to abstain from voting altogether. There are four outcomes that are relevant for the voter:

$$\begin{aligned} a_1 &= \text{abstains, } A \text{ wins} \\ a_2 &= \text{votes, } A \text{ wins} \\ a_3 &= \text{abstains, } B \text{ wins} \\ a_4 &= \text{votes, } B \text{ wins.} \end{aligned}$$

The election is majority-rule, and we suppose a fair coin is flipped in case of a tie. We assume that, from the voter's point of view, the actions of other voters are subject to uncertainty. Let v_A denote the number of votes received by A from other voters, and let v_B denote the number of votes received by B from other voters. Let $P(v_A = v_B)$ denote the probability of the event that A and B tie among other voters, with similar notation used for other events. Each of the voter's three possible decisions determines a lottery over A , e.g., if the voter votes for A , then the probability of a_1 is zero, the probability of a_2 is

$$P(v_A \geq v_B) + (1/2)P(v_A = v_B - 1),$$

the probability of a_3 is zero, and the probability of a_4 is

$$P(v_A < v_B - 1) + (1/2)P(v_A = v_B - 1).$$

You should calculate the lotteries induced by voting for B and abstaining.

We assume for simplicity that the voter's preferences for lotteries over these outcomes are represented by a von Neumann-Morgenstern representation u such that

$$\begin{aligned} u(a_1) &= u_A \\ u(a_2) &= u_A + (D - C) \\ u(a_3) &= u_B \\ u(a_4) &= u_B + (D - C), \end{aligned}$$

where u_A is the utility from A winning, $D \geq 0$ is a utility the voter receives from fulfilling his/her responsibilities as a citizen (the "duty term"), and $C \geq 0$ is a cost of voting. Assume for simplicity that $u_A > u_B$. The expected utility from voting for A is then

$$\begin{aligned} EU[\text{vote } A] &= P(v_A \geq v_B)u_A + P(v_A = v_B - 1) \left(\frac{u_A + u_B}{2} \right) \\ &\quad + P(v_A < v_B - 1)u_B + (D - C), \end{aligned}$$

and the expected utility of abstaining is

$$\begin{aligned} EU[\text{abstain}] &= P(v_A \geq v_B + 1)u_A + P(v_A = v_B) \left(\frac{u_A + u_B}{2} \right) \\ &\quad + P(v_A < v_B)u_B. \end{aligned}$$

You can check that

$$\begin{aligned}
& EU[\text{vote A}] - EU[\text{abstain}] \\
&= P(v_A = v_B)u_A + \left(\frac{u_A + u_B}{2}\right) (P(v_A = v_B - 1) - P(v_A = v_B)) \\
&\quad - P(v_A = v_B - 1)u_B + (D - C) \\
&= P(v_A = v_B) \left(\frac{u_A - u_B}{2}\right) + P(v_A = v_B - 1) \left(\frac{u_A - u_B}{2}\right) + (D - C) \\
&= P(v_A = v_B \text{ or } = v_B - 1) \left(\frac{u_A - u_B}{2}\right) + (D - C).
\end{aligned}$$

Note that $P(v_A = v_B \text{ or } = v_B - 1)$ is just the probability that the voter's vote will make a difference in the election, or the probability of being "pivotal." In large elections, we would expect this to be *very* close to zero.

The term "paradox of voting" refers to the belief, held by some, that $D = 0$, that $C > 0$, and that $u_A - u_B$ is not tremendously large compared to C . In this case, the above calculations suggest that the voter should abstain. If this were true for all voters, then we would get the strange conclusion that no one votes. Of course, this is entirely an artifact of the premise that $D - C < 0$; if you allow for the possibility that $D - C \geq 0$ for some voters, then they will indeed vote.

Going back to the strange conclusion of the paradox of voting, notice that no one votes, so the candidates would tie, and every voter could decide the election by voting. That is, the probability of being pivotal would be *one* for each voter. But then every voter would want to vote, but then the probability of being pivotal would be (essentially) *zero* again, and no voter would be pivotal, and so on. This points to a difficulty with assuming the probability distributions for the random variables v_A and v_B are exogenous, and it gives an example of strategic interdependence, where what one person wants to do depends on his/her expectations about what others will do. This is the subject matter of game theory.

References

- [1] J. von Neumann and O. Morgenstern (1944) *Theory of Games and Economic Behavior*, Princeton: Princeton University Press.